

# **Anna University**

## **REGULAR EXAMINATION ? 2023**

Electrical Machines-I ? Regular Examination, 2023

Subject Code: EE301 Semester: 3 Regulation: R23

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.  
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### **SECTION A (Unit 1: Magnetic Circuits)**

1. How is Magnetic Circuits implemented? Give an algorithm / procedure and explain. [12 Marks]
2. Define Magnetic Circuits. Explain its significance in Electrical Machines-I with a suitable example. [12 Marks]
3. Describe the working principle of Magnetic Circuits with necessary equations. [12 Marks]

### **SECTION B (Unit 2: Transformers)**

4. Solve a problem related to Transformers. Show all the steps clearly. [12 Marks]
5. Explain Transformers with a neat diagram/sketch and a suitable example. [12 Marks]
6. Compare and contrast the key concepts involved in Transformers. [12 Marks]

### **SECTION C (Unit 3: DC Generators)**

7. Compare and contrast the key concepts involved in DC Generators. [12 Marks]
8. Derive the expression related to DC Generators and explain each term. [12 Marks]
9. How is DC Generators implemented? Give an algorithm / procedure and explain. [12 Marks]

### **SECTION D (Unit 4: DC Motors)**

10. Write short notes on DC Motors. [12 Marks]
11. List the advantages and limitations of DC Motors. Explain any two in detail. [12 Marks]
12. Solve a problem related to DC Motors. Show all the steps clearly. [12 Marks]

### **SECTION E (Unit 5: Testing of DC Machines)**

13. List the advantages and limitations of Testing of DC Machines. Explain any two in detail. [12 Marks]
14. Write short notes on Testing of DC Machines. [8 Marks]
15. Derive the expression related to Testing of DC Machines and explain each term. [12 Marks]

\*\*\* End of Question Paper \*\*\*